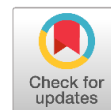


Research Article



Potential of botanical biostimulants (peppermint, laurel, and rosemary extracts) to influence growth, physiological performance, antioxidant defense, and postharvest quality of hydroponically grown tuberose (*Polianthes tuberosa* L.): a preliminary investigation

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ABSTRACT

This study evaluated the effectiveness of botanical biostimulants in enhancing the growth, physiological performance, and postharvest quality of hydroponically grown tuberose (*Polianthes tuberosa* L.) cut flowers. Two nutrient solutions were compared: the conventional Hoagland nutrient solution (Solution A) and a modified nutrient solution supplemented with peppermint, laurel, and rosemary leaf extracts (Solution B). Plant growth, floral characteristics, vase life, biochemical traits, and antioxidant enzyme activities were assessed under both treatments. The results demonstrated that the botanical biostimulants significantly improved several growth and postharvest attributes compared with the conventional Hoagland solution. Peppermint extract at 15 mL L⁻¹ accelerated

KEYWORDS

Correlation coefficients;
Hydroponic systems;
Plant nutrition;
Principal component analysis;
Tuberose;
Medicinal plant extracts;
Vase life

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flower emergence (96.3 days), increased reducing sugar content (0.218), and induced the highest activities of peroxidase and catalase. Rosemary extract at 15 mL L⁻¹ significantly extended vase life (8.33 days), while rosemary extract at 5 mL L⁻¹ produced the highest number of florets (27.67). Laurel extract at 10 mL L⁻¹ significantly increased spike length (26.83 cm). In contrast, the control treatment (Hoagland solution) produced the highest number of bulblets. Pearson's correlation analysis revealed significant positive relationships among several growth and physiological traits. Leaf number was positively correlated with flower diameter ($r = 0.373^*$), spike length ($r = 0.412^*$), and fresh weight ($r = 0.747^{**}$). Flower diameter showed positive correlations with fresh weight ($r = 0.508^{**}$), peroxidase activity ($r = 0.419^*$), and catalase activity ($r = 0.419^*$). Flower number was positively associated with spike length ($r = 0.511^{**}$), whereas peduncle length and bulblet number were positively correlated with reducing sugar content ($r = 0.494^{**}$ and $r = 0.368^*$, respectively). A perfect positive correlation was observed between peroxidase and catalase activities ($r = 1.000^{**}$). Principal component analysis identified five principal components explaining 76.86% of the total variation. Overall, the findings indicate that peppermint, rosemary, and laurel leaf extracts can serve as effective botanical biostimulants for improving the growth, physiological traits, and postharvest performance of hydroponically grown tuberose, highlighting their potential as sustainable supplements to conventional mineral fertilization.

1. Introduction

Tuberose (*Polianthes tuberosa* L.) is a perennial bulbous herb belonging to the family Agavaceae within the monocotyledonous group. Owing to its attractive appearance, pleasant fragrance, and excellent postharvest characteristics, it is widely cultivated as an ornamental cut flower (Babarabie et al., 2026). In Iran, the cultivation of tuberose has expanded considerably due to favorable climatic conditions that support its growth, together with access to suitable export markets, which has increased its commercial importance (Salehi Sardoei et al., 2023). Furthermore, Iran's strategic geographical location and proximity to major consumer markets provide favorable opportunities for the development of the ornamental horticulture and cut flower industries.

In recent years, hydroponic cultivation has been increasingly adopted for the production of ornamental crops, particularly cut flowers such as tuberose (Sweta Kumari et al., 2018). This cultivation system is especially advantageous in regions facing water scarcity, where hydroponics has emerged as one of the most efficient approaches for improving water-use efficiency in agricultural production (Pomoni et al., 2023). Supraja et al. (2020) reported that the daily water consumption of most soilless cultivation systems ranges from 25.0 to 25.1 L m⁻² of growing area. In addition to water conservation, soilless cultivation offers several advantages over conventional soil-based systems, including precise control of nutrient supply, reduced incidence of soil-borne diseases and pests, and improved yield, product quality, and production efficiency (Pomoni et al., 2023; Supraja et al., 2020).

The application of nutrient solutions in hydroponic systems has been extensively investigated for a wide range of crops. However, limited information is available regarding nutrient formulations specifically developed for ornamental cut flowers such as tuberose. Optimizing nutrient solutions for this crop may improve vegetative growth, flowering performance, vase life, and the accumulation of valuable bioactive compounds. Although numerous nutrient formulations have been proposed to enhance the growth and productivity of different plant species, relatively few studies have focused

on developing specialized nutrient solutions for cut flowers (Gómez et al., 2022) .

Medicinal plants such as peppermint (*Mentha × piperita* L.), rosemary (*Rosmarinus officinalis* L.), and laurel (*Laurus nobilis* L.) are rich sources of bioactive compounds, including flavonoids, tannins, phenolic acids, saponins, and vitamins, which are predominantly concentrated in their leaves. These phytochemicals confer strong antioxidant properties and contribute to improved plant physiological performance (Gómez et al., 2022; Sataie-Mokhtari et al., 2026; Salehi Sardoei et al., 2026) . In addition, their essential oils possess well-documented antimicrobial activity, making them promising candidates for use as natural biostimulants in sustainable crop production systems (Babarabie et al., 2025; Ghorbanpour et al., 2025) .

Therefore, the present study aimed to compare the effects of the conventional Hoagland nutrient solution with those of a nutrient solution supplemented with peppermint, laurel, and rosemary extracts as botanical biostimulants on the growth, flowering, bulb formation, and postharvest vase life of hydroponically grown tuberose cut flowers.

2. Materials and methods

2.1. Plant materials and growth conditions

Uniform pearl tuberose (*Polianthes tuberosa* L.) bulbs, each approximately 2 cm in diameter and weighing about 45 g, were obtained from the Tuberose Flower Cultivation Center in Dezful, southwestern Iran. Two bulbs were planted in each 25-cm-diameter plastic pot containing a 1:1:1 (v/v/v) mixture of peat moss, sand, and clay soil as the growing medium. A 1–2 cm layer of mineral perlite was placed at the bottom of each pot to improve drainage. Prior to planting, the bulbs were surface-sterilized by immersion in a 2 g L⁻¹ Benomyl fungicide solution for 20 min to minimize the risk of fungal infection and ensure uniform plant establishment.

2.2. Preparation of rosemary, peppermint, and laurel extracts

Dried rosemary, peppermint, and laurel plant materials were obtained from Noavaran Mehr Golestan Company (Gorgan, Iran). According to the supplier, the plants were harvested before flowering and subsequently dried using industrial drying systems.

Plant extracts were prepared by soaking 100 g of dried plant material in 1 L of distilled water for 24 h. The mixtures were then filtered to obtain the aqueous extracts according to the method described by Iqbal et al., (2011) . The chemical composition of the leaf extracts is presented in **Table 1**. Elemental analysis of the extracts was performed by Dr. Moghasri's Soil and Water Laboratory (Gorgan, Iran), and the analytical reports were provided to the authors.

Table 1. Composition of rosemary, peppermint and laurel extracts in this study.

Trait	Rosemary	Peppermint	Laurel
N (%)	1.63	1.93	1.33
P (%)	0.3	0.5	0.6
K (%)	2.3	2.90	1.80
Ca (%)	2	3.62	2.90
Mg (%)	0.6	0.34	1.82
Cu (ppm)	13.2	13.30	11.20
Fe (ppm)	535	893	222
Mn (ppm)	18.9	30.94	19.98
Zn (ppm)	21.64	40.12	14.64
B (ppm)	9.57	66	47.10
Dry matter (%)	94.95	94.70	95.60

2.3. Treatments

The experiment included aqueous extracts of peppermint, rosemary, and laurel applied at concentrations of 5, 10, and 15 mL L⁻¹. These concentrations were selected based on preliminary experiments and previous studies (Koriesh et al., 2018). Stock extracts were diluted with water to prepare the required concentrations before application.

The extract solutions were applied to the root zone through irrigation every three days throughout the experimental period. The control plants received the standard Hoagland nutrient solution at the same application interval, and its composition is presented in **Table 2** (Resh 2012). Treatments continued for approximately six months, from planting until the completion of the flowering period.

Table 2. The composition, electrical conductivity (EC), salt index (SI), pH, and the mineral content of nutrient solutions used in this study.

EC	1.64
pH	6.9
SI	13.1
Salts	mg/L
KNO ₃	505
Ca (NO ₃) ₂	1180
NH ₄ NO ₃	80
KH ₂ PO ₄	143
MgSO ₄	493
Na ₂ MoO ₄	0.012
H ₃ BO ₃	2.86
CuSO ₄	0.051
ZnSO ₄	0.22
Fe-EDDHA	22.5
MnCl ₂	1.81

2.4. Measurements

Three inflorescences were randomly selected from each experimental unit when the flower stalk was fully developed and the two basal florets had opened. The cut flowers were transferred to the laboratory and placed in bottles containing distilled water. Laboratory temperature was maintained at 18–22°C throughout the vase-life evaluation.

2.4.1. Growth parameters

Days to flower emergence were calculated as the number of days from bulb planting to peduncle emergence. Peduncle and floret diameters were measured using a digital caliper. Spike and peduncle lengths were measured with a ruler, while inflorescence fresh weight was determined using a digital balance.

2.4.2. Enzyme extraction

Leaf samples (0.5 g) were frozen in liquid nitrogen and ground to a fine powder using a chilled mortar and pestle. The powder was homogenized in 2 mL of extraction buffer containing 50 mM potassium phosphate buffer (pH 8.0), 10% (w/v) polyvinylpyrrolidone (PVP), 0.1 mM ethylenediaminetetraacetic acid (EDTA), and 1 mM dithiothreitol (DTT). The homogenate

was centrifuged at $15,000 \times g$ for 30 min at 4°C , and the resulting supernatant was collected for enzyme assays.

2.4.3. Catalase (CAT)

Catalase (CAT) activity was determined spectrophotometrically according to the method of Aebi (1984) by monitoring the decrease in absorbance at 240 nm resulting from H_2O_2 decomposition. The reaction mixture consisted of 1 mL of 50 mM potassium phosphate buffer (pH 7.0) containing 15 mM H_2O_2 . The reaction was initiated by adding 50 μL of crude enzyme extract. Catalase activity was expressed as $\mu\text{mol H}_2\text{O}_2$ decomposed $\text{min}^{-1} \text{mg}^{-1}$ protein.

2.4.4. Guaiacol peroxidase (POD)

Guaiacol peroxidase (POD) activity was measured by mixing 50 μL of crude enzyme extract with 2 mL of reaction solution containing 50 mM potassium phosphate buffer (pH 7.0), 13 mM guaiacol, and 5 mM H_2O_2 . The increase in absorbance due to guaiacol oxidation (extinction coefficient = $26.6 \text{ mM}^{-1} \text{ cm}^{-1}$) was monitored at 470 nm for 1 min. POD activity was expressed as $\mu\text{mol guaiacol oxidized min}^{-1} \text{mg}^{-1}$ protein (Chance and Maehly 1955).

2.4.5. Flower longevity

Flower longevity was evaluated by visual assessment based on petal discoloration, floret abscission, degree of flower opening, and browning of the cut stem base (Mohammadi et al., 2014). Vase life was considered terminated when approximately 50% of the florets exhibited browning and/or abscission symptoms.

2.4.6. Reducing sugars

Petal tissue (0.02 g) was homogenized in 10 mL of distilled water using a porcelain mortar. The homogenate was transferred to a beaker and heated to boiling, after which it was filtered through Whatman No. 1 filter paper. Two milliliters of the filtrate were mixed with 2 mL of copper sulfate solution, sealed with cotton plugs, and heated in a boiling water bath (100°C) for 20 min. During this process, Cu^{2+} ions were reduced to Cu_2O by reducing sugars, producing a reddish precipitate. After cooling, 2 mL of phosphomolybdic acid solution was added, resulting in the development of a blue color. The solution was mixed thoroughly, and reducing sugar concentration was determined spectrophotometrically at 600 nm (Somogyi 1952).

2.5. Statistical analysis

This experiment was conducted in a completely randomized design (CRD) with three replications ($n = 3$). Data were analyzed using SAS software (version 9.4). First, the normality of the data distribution was confirmed using the Shapiro-Wilk test and the homogeneity of variances using Levene's test. Then, one-way analysis of variance (ANOVA) was performed to determine the significant effect of treatments on the measured traits. If the effect of treatments was significant at the 5% probability level ($p < 0.05$), the means were compared using Duncan's Multiple Range Test (DMRT) at the 5% probability level. Duncan's test was chosen due to the exploratory nature of the study, the number of treatments, and its widespread use in similar research in the field of ornamental plants.

3. Results

3.1. LC-MS Analysis of the chemical composition of peppermint extract

The chemical composition of the peppermint extract was characterized by LC-MS analysis in both positive and negative ionization modes. The identified compounds are presented in **Table 3**. In the negative ionization mode, several phenolic compounds were identified, including rosmarinic acid, caffeic acid, cinnamic acid, and ferulic acid. In the positive ionization

mode, the identified compounds included 3-octanol, eriocitrin, (+)-iso-mintlactone, 1-tridecene, catechin, p-coumaric acid, coumarin, vanillin, 1,8-cineole, neomenthol, trans-ferulic acid, menthol acetate, isocaryophyllene, caryophyllene oxide, and various monoterpenes (α -pinene, β -pinene, limonene, carvone, menthol derivatives) (**Table 3**).

Table 3. Chemical composition of peppermint extract.

Compound	MW	Positive Mode	Negative Mode	Type of Composition	Chemical Formula
Cinnamic Acid	148.16		+	Carboxylic acid	C ₉ H ₈ O ₂
3-Octanol	130.23	+	+	Fatty alcohol	C ₈ H ₁₈ O
Cis-Carane	138.24		+	Terpenoid	C ₁₀ H ₁₈
Caffeic Acid	180.16		+	Cinnamic acid	C ₉ H ₈ O ₄
Rosmarinic Acid	360.32		+	Cinnamic acid–polyphenol	C ₁₈ H ₁₆ O ₈
Eriocitrin	596.5	+	+	Flavonoid-glycoside	C ₂₇ H ₃₂ O ₁₅
Naringenin-7-O-glucoside	434.4		+	Flavonoid-glycoside	C ₂₁ H ₂₂ O ₁₀
3,6-Dimethyl-4,5,6,7-tetrahydro-benzo[b]-furan-2(3H)-one Or (-)-Mintlactone Or (+)-iso-Mintlactone	166.22	+	+	Terpenoid	C ₉ H ₁₄ O ₂
7a-Hydroxy mintlactone Or 1-tridecene	182.22	+		Terpenoid	C ₁₃ H ₂₄ O
Catechin	290.26	+	+	Polyphenol	C ₁₅ H ₁₄ O ₆
p-Coumaric acid	160.16	+	+	Cinnamic acid	C ₉ H ₈ O ₃
Coumarin	146.14	+	+	Coumarins	C ₉ H ₆ O ₂
Carvacrol Or Verbenone Or Menthofuran Or Carvone Or Chrysanthenone	150.22	+	+	Terpenoid	C ₁₀ H ₁₄ O
Vanilin	152.15	+	+	Phenolic aldehyde	C ₈ H ₈ O ₃
1,8-Cineole Or g-Terpineol Or trans-Sabinene hydrate Or Linalool Or Menthone Or p-Menthone Or Isomenthone Or α -Terpineol	154	+	+	Terpenoid	C ₁₀ H ₁₈ O
Neomenthol Or Neoisomenthol Or Menthol Or Isomenthol Or Neomenthol acetate	156	+	+	Terpenoid	C ₁₀ H ₂₀ O
trans-Ferulic acid	194.18	+	+	Cinnamic acid	C ₁₀ H ₁₀ O ₄
Menthol acetate Or Isomenthol acetate	198	+	+	Terpenoid	C ₁₂ H ₂₂ O ₂
Isocaryophyllene Or Cadinene Or β -Bourbonene Or β -Elemene Or b-Caryophyllene Or (Z)- β -farnesene Or α -Humulene Or Germacrene D Or Germacrene A	204	+		Terpenoid	C ₁₅ H ₂₄
Caryophyllene oxide	220	+	+	Sesquiterpene	C ₁₅ H ₂₄ O
Viridiflorol	222	+		Sesquiterpene	C ₁₅ H ₂₆ O
Methyl isoheptadecanoate	284	+	+	Fatty ester	C ₁₈ H ₃₆ O ₂
α -Thujene Or α -Pinene Or b-Pinene Or α -Phellandrene Or β -Phellandrene Or α -Terpinene Or g-Terpinene Or Sabinene Or Myrcene Or (Z)- β -ocimene Or Camphene Or Limonene	136	+		Terpenoid	C ₁₀ H ₁₆

3.2. Identification of phytochemical constituents by LC–MS

LC–MS analysis revealed the presence of a diverse range of phytochemical constituents in the peppermint extract. In the negative ionization mode, several phenolic compounds were identified at different retention times (RTs). In the positive ionization mode, the identified compounds included 3-octanol, eriocitrin, (+)-iso-mintlactone, 1-tridecene, catechin, *p*-coumaric acid, coumarin, vanillin, 1,8-cineole, neomenthol, trans-ferulic acid, menthol acetate, isocaryophyllene, caryophyllene oxide, methyl isoheptadecanoate, α -thujene, α -pinene, β -pinene, α -terpinene, sabinene, myrcene, camphene, and limonene (**Table 3**).

Rosemary contains a diverse array of well-known bioactive compounds, several of which are summarized in **Table 4**. Liquid chromatography-mass spectrometry (LC-MS) analysis in both positive and negative ionization modes confirmed the presence of specific phytochemicals within the phenolic and terpenoid composition of the studied extract. Consistent with the findings for the peppermint extract, overlapping molecular weights were observed among the detected compounds. In positive ion mode, a range of compounds was identified in the rosemary extract, including limonene, α -pinene, 3-carene, camphenol, exo-fenchol, cinnamic acid, retinol, 3-cyclohexene-1-methanol, β -caryophyllene, luteolin, rosmaquinone, rosmanol, rosmadial, and carnosol. A similar profile of phenolic compounds was also identified in negative ion mode (**Table 4**).

Table 4. Chemical composition of rosemary extract.

Natural Compound	MW	Positive Mode	Negative Mode	Type of Composition	Chemical Formula
Limonene Or α -pinene Or 3-carene	136.24	+	+	Terpenoid	C10H16
6-camphenol Or Camphor	152.24	+		Terpenoid	C10H16O
Exo-fenchol	154.25	+		Cannabinoid	C10H18O
Cinnamic acid	148.16		+	Cinnamic acid	C9H8O2
Retinol	286.46	+		Vitamin A	C20H30O
3-cyclohexene-1-methanol	112.17	+			C7H12O
β -caryophyllene	204.36	+		Sesquiterpene	C15H24
Luteolin	286.24	+	+	Flavonoid	C15H10O6
Rosmaquinone	358.43	+	+	Terpenoid	C20H30O4
Rosmanol	346.24	+	+	Polyphenol	C20H28O4
Rosmadial	344.41	+	+	Terpenoid	C20H28O3
Carnosol	330.42	+	+	Terpenoid	C20H28O4
Ursolic acid	456.71		+	Terpenoid-bile acid	C30H48O3
Rosmarinic acid	360.32		+	Cinnamic acid-polyphenol	C18H16O8
Oleanolic acid	456.71		+	Terpenoid-bile acid	C30H48O3
Carnosic acid	332.44		+	Terpenoid	C20H28O4
Caffeic acid	180.16		+	Cinnamic acid	C9H8O4

Table 5 presents the phytochemical composition of the laurel (*Laurus nobilis*) extract. A total of seventeen major constituents were identified in the sample. In positive ion mode, the detected compounds included α -pinene, (+)-sabinene, carvone, eucalyptol, 3,7-dimethylocta-1,6-dien-3-ol, β -fenchol, terpineol, eugenol, caryophyllene, germacra-1(10),4(15),5-triene, cis- α -bisabolene, caryophyllene oxide, (–)-spathulenol, gazaniolide, spirafolide, and phytol. Corresponding compounds were also identified in negative ion mode (**Table 5**).

Table 5. Chemical composition of laurel extract.

Natural Compound	MW	Positive Mode	Negative Mode	Type of Composition	Chemical Formula
Cumene	120	+		Terpenoid	C ₉ H ₁₂
α -Pinene / (+)-Sabinene	136	+	+	Terpenoid	C ₁₀ H ₁₆
Carvone	150	+	+	Terpenoid	C ₁₀ H ₁₄ O
Eucalyptol / 3,7-Dimethylocta-1,6-dien-3-ol / Beta-fenchol / Terpineol	154	+	+	Terpenoid Cannabinoid	C ₁₀ H ₁₈ O
Eugenol	164		+	Cinnamic acid	C ₁₀ H ₁₂ O ₂
Caryophyllene / Germacra-1(10),4(15),5-triene / cis- α -Bisabolene	204	+	+	Sesquiterpene	C ₁₅ H ₂₄
Caryophyllene oxide / (-)-Spathulenol	220	+	+	Sesquiterpene	C ₁₅ H ₂₄ O
Gazaniolide	230	+	+	Sesquiterpene	C ₁₅ H ₂₂ O ₃
Spirafolide	246	+	+	Sesquiterpene lactone Flavonoid	C ₁₅ H ₂₂ O ₃
Phytol	296		+	Terpenoid	C ₂₀ H ₄₀ O

3.3. Number of leaves

Based on Duncan's Multiple Range Test at the 5% significance level, the highest number of leaves (10.33) was recorded in the 10 mL L⁻¹ rosemary extract treatment, which was not significantly different from the 5 mL L⁻¹ laurel extract treatment (10.33) and the 5 mL L⁻¹ rosemary extract treatment (10.00). The lowest number of leaves (6.33) was observed in the 15 mL L⁻¹ rosemary extract treatment (**Table 6**).

3.4. Peduncle and floret diameter

The maximum peduncle diameter (7.03 mm) was recorded in the 5 mL L⁻¹ laurel extract treatment, which was not significantly different from the 10 mL L⁻¹ peppermint extract treatment (6.85 mm) and the control treatment (6.60 mm). The minimum peduncle diameter (5.16 mm) was observed in the 15 mL L⁻¹ rosemary extract treatment. The maximum floret diameter (49.51 mm) was recorded in the control treatment, which was not significantly different from the 5 mL L⁻¹ laurel extract treatment (49.92 mm) and the 5 mL L⁻¹ peppermint extract treatment (49.65 mm) (**Table 6**).

3.5. Floret count

The maximum floret count (27.67) was observed in samples treated with 5 mL L⁻¹ rosemary extract. In contrast, the lowest floret count (13.33) was recorded following the combined application of peppermint and laurel extracts (15 mL L⁻¹ each) (**Table 6**).

3.6. Length of peduncle and spike

The maximum peduncle length (82.67 cm) was recorded following the application of 5 mL L⁻¹ peppermint extract, which was not statistically different from the 5 and 10 mL L⁻¹ rosemary extract treatments. Conversely, the shortest peduncles (52.50 cm) were observed in the 15 mL L⁻¹ rosemary extract treatment (**Table 6**). For spike length, the maximum value (26.83 cm) was achieved when tuberosc bulbs were treated with 10 mL L⁻¹ laurel extract, whereas the minimum length (17.33 cm) was recorded with 15 mL L⁻¹ rosemary extract. No significant differences in spike length were found

among the treatments with 15 mL L⁻¹ laurel extract, 5 and 10 mL L⁻¹ peppermint extract, and 10 mL L⁻¹ rosemary extract (**Table 6**).

Table 6. Effects of plant extracts (Rosemary, Laurel, Peppermint) on morphological characteristics of cut Tuberose flowers.

Treatments	Number of leaves	Peduncle diameter (mm)	Floret diameter (mm)	Number of florets	Peduncle length (cm)	Spike length (cm)
Rosemary (5ml/l)	9 bc	5.85 bc	42.46 cd	27.66 a	79.33 a	25.33 ab
Rosemary (10ml/l)	10.33 a	6.51 ab	43.38 bcd	24.00 ab	76.66 a	19 c
Rosemary (15ml/l)	6.33d	5.16 c	32.71 e	22.33 abc	52.500 b	17.33 c
Peppermint (5ml/l)	9 bc	6.69 ab	49.650 a	17.66 abc	82.66 a	22.50 bc
Peppermint (10ml/l)	7.66 bcd	6.85 ab	44.950 abc	15.66 bc	76.66 a	18.50c
Peppermint (15ml/l)	7 cd	5.85 bc	42.750 cd	13.33 c	71.33 a	18 c
Laurel (5ml/l)	10.33 a	7.03 a	49.92 a	20.66 abc	81.33 a	25.83 ab
Laurel (10ml/l)	9 ab	6.35 ab	48.25 ab	24.33 ab	8.50 a	26.83 a
Laurel (15ml/l)	8.33 bc	6.07 abc	38.75 d	13.33 c	78.33 a	17.66 c
Control (Hoagland)	9 ab	6.60 ab	49.50 a	18.00 abc	81.33 a	20.83 bc

Notes: †Means in each column with the same letters are not statistically significantly different at $p < 0.05$ based on Duncan's multiple range test.

3.7. Fresh weight of flowers

The maximum fresh flower weight (103.29 g) was recorded in samples treated with 5 mL L⁻¹ laurel extract. Conversely, the minimum fresh weight (55.32 g) was observed in the 15 mL L⁻¹ rosemary extract treatment, which was not statistically different from the 15 mL L⁻¹ peppermint extract treatment (**Table 7**).

3.8. Number of bulblets

The control treatment (Hoagland solution) yielded the highest number of bulblets (16.00). In contrast, the lowest bulblet count (8.33) was recorded in samples treated with a combination of 15 mL L⁻¹ peppermint and 15 mL L⁻¹ laurel extracts (**Table 7**).

3.9. Flower emergence

The longest duration to flower emergence (96.33 days) was observed in samples treated with 15 mL L⁻¹ peppermint extract. Conversely, the shortest emergence period (75.66 days) occurred with the 5 mL L⁻¹ rosemary extract treatment, which differed significantly from the control group (**Table 7**).

3.10. Reducing sugars

The minimum reducing sugar content (0.218 mg g⁻¹ FW) was recorded in the 15 mL L⁻¹ peppermint extract treatment, which was not significantly different from the 5 mL L⁻¹ rosemary extract treatment. The maximum reducing sugar

concentration (0.520 mg g⁻¹ FW) was observed in the control group (**Table 7**).

3.11. Peroxidase activity

Peroxidase (POD) activity reached its maximum in samples treated with 15 mL L⁻¹ peppermint extract. Conversely, the minimum POD activity was recorded in the 15 mL L⁻¹ laurel extract treatment (**Table 7**).

3.12. Catalase activity

Catalase (CAT) activity was highest following the application of 15 mL L⁻¹ peppermint extract, whereas the lowest CAT activity was observed in samples treated with 15 mL L⁻¹ laurel extract (**Table 7**).

3.13. Vase life

The maximum vase life (8.33 days) was recorded in samples treated with 15 mL L⁻¹ rosemary extract, while the minimum vase life (6.33 days) was observed in the 15 mL L⁻¹ peppermint extract treatment (**Table 7**).

Table 7. Effects of plant extracts (rosemary, laurel, peppermint) on morphological and physiological characteristics of cut Tuberose flowers.

Treatments	Fresh weight (g)	Number of bulblets	Flower emergence (days)	Reducing sugars (mg/g FW)	POD (U/mg protein)	CAT (U/mg protein)	Vase life (days)
Rosemary (5ml/l)	92.25 b	12.66 c	75.66 g	0.21 g	0.97 e	0.74e	7.66 ab
Rosemary (10ml/l)	64.46 cd	10.66 de	84 ef	0.41 d	1.37 b	1.11 b	8 a
Rosemary (15ml/l)	55.32 d	10 e	90.66 bc	0.39 e	0.92 e	0.70e	8.33 a
Peppermint (5ml/l)	82.46 b	13.66 b	91.66 bc	0.44 b	1.06 d	0.83 d	8 a
Peppermint (10ml/l)	58.13 cd	10 e	93 ab	0.31 f	1.37 b	1.10 b	7.3 bc
Peppermint (15ml/l)	55.86 d	8.33 f	96.33 a	0.21 g	1.46 a	18 a	0.33 d
Laurel (5ml/l)	103.29 a	14.33 b	82.33 f	0.45 b	1.28 c	1.02 c	7.66 abc
Laurel (10ml/l)	85.83 b	11 d	89 cd	0.42 c	0.94 e	0.72 e	7.66 abc
Laurel (15ml/l)	61.93 cd	8.33 f	93.33 ab	0.41 cd	0.84 f	0.63 f	7 cd
Control (Hoagland)	67.78 c	16 a	86.33 de	0.52 a	0.97 e	0.74 e	0.33 d

Notes: † Means in each column with the same letters are not statistically significantly different at $p < 0.05$ based on Duncan's multiple range test.

3.14. Correlation coefficient analysis

Pearson correlation analysis revealed significant positive relationships among several morphological and biochemical

traits (**Figure 1**). The correlation coefficients are reported as follows:

- Leaf number was positively correlated with floret diameter ($r = 0.373$, $p < 0.05$), spike length ($r = 0.412$, $p < 0.05$), and fresh weight ($r = 0.747^{**}$, $p < 0.01$).
- Floret diameter showed positive correlations with fresh weight ($r = 0.508^{**}$, $p < 0.01$), peroxidase activity ($r = 0.419$, $p < 0.05$), and catalase activity ($r = 0.419$, $p < 0.05$).
- Floret number was positively associated with spike length ($r = 0.511^{**}$, $p < 0.01$).
- Peduncle length showed a positive correlation with reducing sugar content ($r = 0.494^{**}$, $p < 0.01$).
- Bulblet number was positively correlated with reducing sugar content ($r = 0.368^{*}$, $p < 0.05$).
- Peroxidase and catalase activities exhibited a perfect positive correlation ($r = 1.000^{**}$, $p < 0.01$).

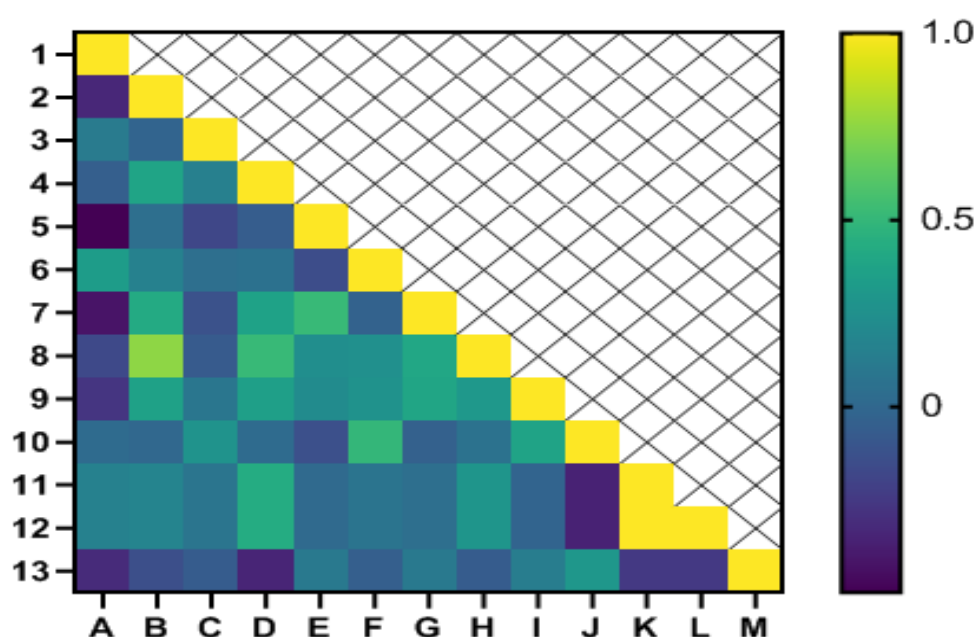


Figure 1. Heat map of mutual relations of variables in correlation coefficient for physicochemical indicators.

Y axis are respectively: **1-** Flowering Time, **2-** N. Leaf, **3-** Stem Diameter, **4-** Flower Diameter, **5-** Number of Flowers, **6-** Stem Length, **7-** Spike Length, **8-** Fresh Weight, **9-** N. Bulbelt, **10-** Reducing Sugars, **11-** Peroxidase, **12-** Catalase, **13-** Vase Life.

X axis are respectively: **A-** Flowering Time, **B-** N. Leaf, **C-** Stem Diameter, **D-** Flower Diameter, **E-** Number of Flowers, **F-** Stem Length, **G-** Spike Length, **H-** Fresh Weight, **I-** N. Bulbelt, **J-** Reducing Sugars, **K-** Peroxidase, **L-** Catalase, **M-** Vase Life.

3.15. Principal component analysis

Prior to PCA, all data were standardized using Z-score transformation (mean = 0, standard deviation = 1) to eliminate the effect of different measurement units (days, cm, mm, g, mg g⁻¹ FW, and enzyme units). PCA was performed based on the correlation matrix, as significant correlations among variables were confirmed in Section 3.14.

The results showed that the first two principal components explained 25.05% and 19.93% of the total variance,

respectively (a cumulative of 44.98%). The first five principal components collectively explained 76.86% of the total variance (**Table 8**).

Based on the eigenvectors:

- **PC1 (25.05%):** Positively associated with vegetative growth traits, including leaf number (0.394), fresh weight (0.308), and floret diameter (0.292).
- **PC2 (19.93%):** Positively associated with antioxidant enzyme activities, including peroxidase (0.484) and catalase (0.484).
- **PC3 (15.58%):** Positively associated with floret number (0.371) and vase life (0.347).
- **PC4 (8.34%):** Positively associated with peduncle length (0.520) and reducing sugars (0.402).
- **PC5 (7.95%):** Positively associated with peduncle diameter (0.761).

Table 8. Eigenvectors, eigenvalues and relative and cumulative variances resulted from principal component analysis of *Polianthes tuberosa* var Pearl.

Traits	PC1	PC2	PC3	PC4	PC5
Flowering time	0.024	0.073	-0.348	0.129	-0.125
N. Leaf	0.394	-0.133	-0.057	-0.036	-0.090
Stem Diameter	-0.052	0.028	0.020	-0.046	0.761
Flower Diameter	0.292	0.016	-0.061	-0.109	0.280
Number of Flowers	-0.106	0.126	0.371	0.010	-0.117
Stem Length	0.016	0.109	-0.146	0.520	-0.270
Spike Length	0.142	-0.013	0.245	-0.030	-0.006
Fresh Weight	0.308	0.003	-0.016	0.097	-0.190
N. Bulbelt	0.080	0.004	0.173	0.232	0.189
Reducing Sugars	-0.034	-0.091	0.008	0.402	0.190
Peroxidase	-0.107	0.484	0.066	0.066	0.021
Catalase	-0.107	0.484	0.066	0.066	0.021
Vase Life	-0.338	0.135	0.347	0.303	-0.017
Eigenvalue	3.257	2.592	2.025	1.084	1.034
Relative Variance	25.055	19.935	15.579	8.341	7.955
Cumulative Variance	25.055	44.989	60.569	68.909	76.864

4. Discussion

The phytochemical analysis of peppermint extract revealed a diverse array of bioactive compounds, including phenolic acids (rosmarinic, caffeic, cinnamic, and ferulic acids), flavonoids (eriocitrin, catechin), and various terpenoids (α -pinene, β -pinene, limonene, carvone, menthol derivatives) (**Table 3**). The presence of glycosidic compounds, such as eriocitrin, which can fragment during ionization, producing multiple mass signals, explains the detection of compounds with similar molecular weights. Notably, peppermint and rosemary extracts share several key compounds, including caryophyllene oxide, β -caryophyllene, rosmarinic acid, cinnamic acid, caffeic acid, and limonene, suggesting potential similarities in their biological activities.

The application of plant extracts as natural biostimulants has gained attention as a sustainable alternative to synthetic growth regulators, which can be costly and potentially harmful when overused (Ahmed et al., 2018; Hamouda et al., 2012).

In the present study, a dose-dependent response was observed, with optimal concentrations (5-10 mL L⁻¹) of the extracts promoting vegetative growth, while higher concentrations (15 mL L⁻¹) exhibited inhibitory effects. For instance, the 10 mL L⁻¹ rosemary treatment resulted in the highest leaf count (10.33), whereas the 15 mL L⁻¹ treatment significantly reduced this parameter to 6.33 (**Table 6**). This biphasic response aligns with previous studies demonstrating that plant extracts can stimulate growth at low doses but become phytotoxic at higher concentrations due to allelopathic compounds (El Sherif 2017).

The growth-promoting effects observed at optimal concentrations can be attributed to several mechanisms. First, plant extracts contain essential micronutrients that supplement the nutrient solution. Ulkuhan et al. (2012) identified essential minerals in laurel leaves and stems, while Arslan and Musa Ozcan (2008) reported the presence of phosphorus, calcium, iron, and zinc in rosemary leaves. Second, these extracts may contain or stimulate the production of plant growth regulators. For example, *Aloe vera* extract has been shown to promote vegetative growth in okra and hibiscus (El Sherif 2017), and its gibberellic acid content (19 mg per 100 g dry weight) has been linked to increased stem elongation in pine seedlings (Nadia et al., 2013). Similarly, *Moringa oleifera* leaf extract, rich in proteins (5-43%) and growth hormones such as auxin and cytokinin, improved growth parameters in basil and senna plants (Balakumbahan and Rajamani 2010; Moyo 2011).

The enhancement of morphological traits, including increased leaf number, spike length, and floret count, may also result from improved physiological processes. Plant extracts can enhance cell membrane permeability, oxygen uptake, respiration, photosynthesis, and ion transport, thereby promoting cell expansion and division (Padmaja et al., 2007; Foidl et al., 2006). These effects are particularly relevant for flower development, as petal growth and flower opening are driven by cell expansion, which requires adequate water and osmoregulating compounds such as carbohydrates (Knee 2000; Sheikh et al., 2013).

However, the inhibitory effects observed at higher extract concentrations (15 mL L⁻¹) warrant further investigation. The reduction in leaf count, floret number, and other growth parameters at supra-optimal doses may be due to the accumulation of growth-inhibitory secondary metabolites with allelopathic properties. High concentrations of certain terpenoids or phenolic compounds can induce oxidative stress, disrupt membrane integrity, and interfere with normal metabolic pathways. This hypothesis is supported by the observed decrease in reducing sugar content in the 15 mL L⁻¹ peppermint treatment (0.218 mg g⁻¹ FW), suggesting impaired carbohydrate metabolism under stress conditions. Similar inhibitory effects have been reported in other studies where increasing doses of plant extracts led to reduced growth parameters (Farhoudi 2015; Maffei et al., 2009).

The variability in responses across different traits and extract types underscores the complexity of plant extract applications. The effects depend not only on the concentration but also on the specific composition of secondary metabolites, which varies among plant species and even within the same species depending on growing conditions, harvest time, and extraction methods (Ulkuhan et al., 2012; Arslan and Musa Ozcan 2008). Therefore, optimizing the concentration and identifying the most effective extract for each species and trait is crucial for maximizing benefits while avoiding phytotoxicity.

The 15 mL L⁻¹ peppermint extract treatment resulted in the lowest reducing sugar content (0.218 mg g⁻¹ FW), which was not significantly different from the 5 mL L⁻¹ rosemary treatment. This reduction in carbohydrate content, contrary to the growth-promoting effects observed at lower concentrations, suggests that supra-optimal doses of plant extracts may impair carbohydrate metabolism, possibly due to the accumulation of allelopathic compounds or induced oxidative stress. This finding contrasts with studies reporting increased sugar content following foliar application of *Moringa oleifera* extract in rice (Foidl et al., 2006) or *Artemisia* essential oil in chrysanthemum (Samiee et al., 2013), where optimal concentrations enhanced photosynthetic activity and carbohydrate accumulation.

The impact of plant extracts on antioxidant enzyme activities further supports the hypothesis of dose-dependent stress responses. In the present study, the 15 mL L⁻¹ peppermint treatment significantly altered peroxidase (POD) and catalase (CAT) activities. Similar biphasic responses have been reported by Farhoudi (2015), who found that low concentrations

(4%) of *Eucalyptus* extract increased POD and CAT activities in *Xanthium strumarium* seedlings, while higher concentrations (16-20%) led to enzyme suppression. This pattern is consistent with the allelopathic effects of secondary metabolites, which at high concentrations can induce oxidative stress, disrupt cell membrane integrity, and inhibit antioxidant enzyme activities, as demonstrated in cucumber (Maffei et al., 1999) and wild mustard seedlings (Oracz et al., 2007). The decrease in reducing sugars combined with altered antioxidant enzyme activities in the 15 mL L⁻¹ treatments indicates that excessive concentrations of bioactive compounds may overwhelm the plant's defense systems, leading to metabolic impairment and growth retardation.

The significant extension of vase life in the 15 mL L⁻¹ rosemary extract treatment (8.33 days compared to 6.33 days in the control) is a notable finding that may be related to the antioxidant and antimicrobial properties of rosemary-specific compounds (such as carnosic acid, carnosol, and rosmarinic acid - **Table 4**). These compounds may scavenge reactive oxygen species (ROS), delay senescence, and prevent microbial colonization in the vase solution, thereby maintaining xylem functionality and improving water uptake, as suggested by previous studies (Samiee et al., 2013; Admane et al., 2023).

The outcomes of this experiment align with previous studies demonstrating that mineral nutrition significantly impacts the growth of annual plants (Ericsson 1995; EBIC 2016; Bulgari 2015). Essential elements found in medicinal plant extracts, such as rosemary, can enhance growth and development when supplied at optimal concentrations. However, achieving high-quality production requires careful optimization of extract concentrations to ensure balanced nutrient availability, regular flowering, and extended flower longevity. Our study demonstrates that extracts from peppermint, rosemary, and laurel can serve as effective natural additives in soilless cultivation systems.

Leaf size, a critical factor influencing growth rate, is particularly important as larger leaves offer greater surface area for light absorption, thereby promoting photosynthesis and overall plant growth. These findings are consistent with Ericsson (1995), who examined the impact of nutrient availability on young plant growth. Additionally, our study suggests that production capacity can be enhanced by including natural compounds that contribute to the synthesis of beneficial organic substances. These compounds function as biostimulants and elicitors. According to the European Biostimulants Industry Council (2016), plant biostimulants are defined as natural substances and/or microorganisms that, when applied to plants, stimulate natural processes to improve nutrient uptake, nutrient efficiency, abiotic stress tolerance, and product quality. The results of our study (**Table 6** and **Table 7**) are consistent with reports from similar studies (2002), confirming the potential of botanical extracts as sustainable alternatives to synthetic chemical regulators in commercial floriculture.

Pearson correlation analysis revealed significant positive associations among key morphological and physiological traits (**Figure 1**). For instance, leaf number was positively correlated with floret diameter ($r = 0.373^*$), spike length ($r = 0.412^*$), and fresh weight ($r = 0.747^{**}$). These strong inter-trait relationships suggest that treatments promoting vegetative growth concurrently enhance overall plant performance and biomass accumulation, a pattern consistent with findings in other horticultural crops (Abouzari et al., 2020).

Principal Component Analysis (PCA) further elucidated the multivariate relationships among the evaluated traits, effectively reducing data dimensionality. The first five principal components collectively explained 76.84% of the total variance. PC1 was heavily loaded with vegetative growth parameters (leaf number, fresh weight, and floret diameter), while PC2 was primarily associated with antioxidant defense mechanisms (peroxidase and catalase activities). PC3 and PC4 grouped reproductive and postharvest traits (floret count, vase life, spike length, and peduncle length) alongside metabolic indicators (reducing sugars and bulblet number). This clustering confirms that the applied botanical extracts induced coordinated, multidimensional shifts in tuberose physiology, allowing for the identification of key traits for future selection programs (Zahedi et al., 2016; Asgar et al., 2010; Zakizade et al., 2010; Khajeh et al., 2015) [].

The significant improvements in tuberose growth and postharvest quality observed in this study are strongly linked to the specific bioactive profiles of the applied extracts. Peppermint extract, rich in terpenoids such as menthol, menthone, carvone, and 1,8-cineole, is known to modulate antioxidant defense systems. This likely underpins the significant changes in peroxidase and catalase activities observed under peppermint treatments. Furthermore, while phenolic acids like caffeic

and ferulic acid generally support carbohydrate metabolism, the notable reduction in reducing sugars at the highest peppermint concentration (15 mL L^{-1}) indicates that supra-optimal doses of these bioactive compounds may induce mild metabolic stress or allelopathic inhibition, overriding their growth-promoting potential.

Conversely, rosemary extract provided potent antioxidants, including carnosic acid, carnosol, rosmarinic acid, and luteolin. The growth-promoting properties of luteolin may explain the increased floret count at lower rosemary concentrations, while the anti-senescent and antimicrobial properties of carnosic acid and carnosol likely preserved cellular integrity and xylem functionality, directly contributing to the prolonged vase life (8.33 days). Similarly, laurel extract, abundant in 1,8-cineole, eugenol, α -terpineol, and sabinene, likely influenced shoot development through phytohormonal mimicry. Eugenol, in particular, has been associated with auxin-like activity, which may have driven the observed increases in spike length and meristematic activity.

Thus, the terpenoids, phenolic acids, and flavonoids identified across the three botanical extracts act as effective natural biostimulants. Their synergistic bioactivity offers a promising, eco-friendly strategy to optimize morphological development and postharvest longevity in soilless tuberose cultivation, provided that concentrations are carefully calibrated to avoid phytotoxic effects.

5. Conclusion

This study demonstrated that aqueous extracts of peppermint, rosemary, and laurel at optimal concentrations ($5\text{--}10 \text{ mL L}^{-1}$) can significantly influence several vegetative, reproductive, and quality traits of tuberose in a hydroponic system under experimental (pot) conditions. Key findings include: Rosemary extract at 10 mL L^{-1} produced the highest leaf number (10.33). Laurel extract at 5 mL L^{-1} produced the highest fresh weight (103.29 g) and peduncle diameter (7.03 mm). Peppermint extract at 15 mL L^{-1} induced the highest peroxidase ($1.46 \text{ U mg}^{-1} \text{ protein}$) and catalase ($1.18 \text{ U mg}^{-1} \text{ protein}$) activities. Rosemary extract at 15 mL L^{-1} resulted in the longest vase life (8.33 days), which may be related to the antioxidant and antimicrobial properties of rosemary-specific compounds. However, higher concentrations (15 mL L^{-1}) of certain extracts exhibited inhibitory effects on several traits, indicating the need for careful concentration optimization for each extract type and plant species.

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Conflicts of interests

The authors declare that there is no conflict of interest.

Data availability statement

All the data generated or analyzed during the current study were included in the manuscript.

Author contributions

M.B and A.S.S: Conceptualization, Methodology, Investigation, A.S.S, B.J, F.B, and F.C: Software, Formal Analysis, Validation, Writing - Review and preparation of final version.

Declaration of generative AI and AI-assisted technologies

The manuscript and underlying research were developed and authored entirely by the researchers. Any AI-assisted

edits and suggestions were carefully reviewed and verified by the authors, who take full responsibility for the accuracy, integrity, and content of the final manuscript.

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